Retail Customer Insights

End-to-End Data Analytics Project using Python, PostgreSQL & Power BI

Prepared By

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1. Introduction

1.1 Overview

Data analytics has become an essential part of modern business decision-making. Organizations collect large amounts of customer and transaction data every day, but the real value lies in converting this raw information into meaningful insights. By analyzing customer behavior, businesses can identify purchasing patterns, improve customer satisfaction, optimize inventory management, and increase profitability.

This project presents an end-to-end data analytics workflow using Python, PostgreSQL, and Power BI. The project begins with cleaning and preparing the raw dataset, followed by exploratory data analysis to understand customer behavior. SQL is then used to answer important business questions, and Power BI is used to create an interactive dashboard that summarizes the key findings. The project demonstrates how different analytics tools work together to support data-driven business decisions.

📷 **Figure 1:** Preview of the Retail Customer Shopping Dataset

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31

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2. Business Problem Statement

Retail companies often collect customer transaction data from various sources. However, without proper analysis, this data remains underutilized and cannot effectively support business decisions.

The company wants to understand customer purchasing behavior, identify high-performing product categories, analyze customer demographics, evaluate the impact of discounts and subscriptions, and understand preferred payment and shipping methods. By answering these questions, the company can improve customer retention, optimize marketing campaigns, and enhance overall business performance.

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3. Project Objectives

The objectives of this project are:

- To clean and preprocess retail customer shopping data.
- To perform exploratory data analysis using Python.
- To identify customer purchasing trends.
- To answer business-related questions using SQL.
- To build an interactive Power BI dashboard for visualization.
- To generate business insights and recommendations based on the analysis.

4. Dataset Description

The dataset used in this project contains customer shopping information collected from a retail business. It includes demographic details, purchase information, payment methods, subscription status, product categories, shipping preferences, and customer ratings.

Dataset Attributes

Attribute	Description
Customer ID	Unique customer identifier
Age	Age of the customer
Gender	Gender of the customer
Category	Purchased product category
Purchase Amount	Amount spent by the customer
Location	Customer location
Size	Product size
Color	Product color
Season	Season in which the purchase was made
Review Rating	Customer rating
Subscription Status	Whether the customer has a subscription
Shipping Type	Delivery method selected
Payment Method	Method used for payment
Discount Applied	Indicates if a discount was applied
Promo Code Used	Indicates whether a promotional code was used
Previous Purchases	Number of previous purchases
Preferred Payment Method	Customer's preferred payment option
Frequency of Purchases	Frequency of customer purchases

📷 **Figure 2:** Dataset Information

```
[35]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object 
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

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5. Tools & Technologies Used

Tool	Purpose
Python	Data Cleaning and Analysis
Pandas	Data Manipulation
PostgreSQL	SQL-based Business Analysis
Power BI	Interactive Dashboard Development
Git & GitHub	Version Control and Portfolio Management

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6. Project Methodology

The project follows a structured workflow to transform raw customer data into meaningful business insights.

Workflow

Retail Customer Dataset
│
▼
Data Cleaning & Preprocessing (Python)
│
▼
Exploratory Data Analysis (EDA)
│
▼
Business Analysis using PostgreSQL
│
▼
Dashboard Development (Power BI)
│
▼
Business Insights & Recommendations

This workflow ensures that the data is accurate, well-structured, and suitable for business analysis and visualization.

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7. Data Cleaning & Preprocessing

Data cleaning is a crucial step in the analytics process because inaccurate or inconsistent data can lead to misleading results. The dataset was cleaned using Python and the Pandas library.

The following preprocessing steps were performed:

- Imported the dataset into Jupyter Notebook.
- Examined data types and dataset structure.
- Checked for missing values.
- Removed duplicate records.
- Renamed columns where required.
- Standardized the dataset for analysis.
- Prepared the cleaned dataset for PostgreSQL.

📷 **Figure 3:** Missing Value Analysis

```
df.isnull().sum()
```

```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

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8. SQL Business Analysis

Overview

After cleaning the dataset in Python, the processed data was imported into a PostgreSQL database. SQL queries were used to answer business-related questions and extract meaningful insights from the retail customer dataset. The analysis focused on customer spending behavior, product performance, subscription patterns, discounts, shipping preferences, and customer segmentation.

The following business questions were addressed:

8.1 Revenue Generated by Gender

Business Question

What is the total revenue generated by male and female customers?

Purpose

This analysis compares the total purchase amount contributed by different customer groups. Understanding revenue by gender helps businesses design targeted marketing campaigns and customer engagement strategies.

📷 **Figure 5:** SQL Query – Revenue by Gender

	gender text	revenue numeric
1	Female	75191
2	Male	157890

8.2 High-Value Customers Using Discounts

Business Question

Which customers used discounts but still spent more than the average purchase amount?

Purpose

This query identifies customers who remain high-value buyers even after receiving discounts. These customers can be considered for premium loyalty programs and personalized offers.

📷 **Figure 6:** SQL Query – High-Value Discount Customers

customer_id bigint	purchase_amount bigint
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100

8.3 Highest Rated Products

Business Question

Which five products have the highest average customer review ratings?

Purpose

Customer ratings provide valuable feedback regarding product quality and customer satisfaction. Identifying highly rated products helps businesses prioritize inventory and promotional activities.

📷 **Figure 7:** SQL Query – Highest Rated Products

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

8.4 Shipping Method Comparison

Business Question

How does the average purchase amount differ between Standard and Express Shipping?

Purpose

Shipping preferences may influence customer purchasing behavior. This analysis compares customer spending across different delivery methods.

📷 **Figure 8:** SQL Query – Shipping Method Comparison

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

8.5 Subscription Analysis

Business Question

Do subscribed customers spend more than non-subscribed customers?

Purpose

Subscription programs are designed to increase customer loyalty and revenue. This analysis evaluates the effectiveness of subscriptions by comparing average spending and total revenue.

📷 **Figure 9:** SQL Query – Subscription Analysis

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

8.6 Discount Analysis

Business Question

Which five products have the highest percentage of purchases with discounts applied?

Purpose

Understanding which products are frequently purchased with discounts helps businesses evaluate promotional strategies and optimize pricing decisions.

📷 **Figure 10:** SQL Query – Discount Analysis

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

8.7 Customer Segmentation

Business Question

How can customers be categorized based on their previous purchases?

Purpose

Customers were classified into three groups:

- New Customers
- Returning Customers
- Loyal Customers

This segmentation supports customer relationship management and helps businesses create targeted retention strategies.

📷 **Figure 11:** SQL Query – Customer Segmentation

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8.8 Top Products in Each Category

Business Question

Which are the three most purchased products within each product category?

Purpose

This analysis identifies the best-performing products in every category, helping businesses improve inventory planning and product recommendations.

📷 **Figure 12:** SQL Query – Top Products by Category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

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9. Dashboard Development using Power BI

An interactive Power BI dashboard was developed to visualize the results of the analysis. The dashboard provides stakeholders with a clear overview of customer behavior and business performance.

The dashboard includes:

- Key Performance Indicators (KPIs)
- Revenue analysis
- Customer demographics
- Product category performance
- Payment method analysis
- Shipping type analysis
- Interactive filters and slicers

The dashboard enables users to explore the data dynamically and supports informed business decision-making.

📷 **Figure 13:** Retail Customer Insights Dashboard

10. Key Business Insights

The analysis of the retail customer shopping dataset provided several valuable insights into customer behavior, purchasing patterns, and business performance.

1. Revenue Analysis by Gender

The comparison of revenue generated by male and female customers highlighted the contribution of each customer segment to the company's overall sales. This insight can help businesses develop more targeted marketing strategies.

2. High-Value Customers

The analysis identified customers who used discounts but still spent above the average purchase amount. This suggests that promotional offers can effectively encourage higher-value purchases while maintaining customer engagement.

3. Product Performance

The evaluation of customer review ratings helped identify the highest-rated products. Products with consistently high ratings indicate strong customer satisfaction and can be prioritized for marketing and inventory planning.

4. Subscription Impact

The comparison between subscribed and non-subscribed customers provided insights into customer spending patterns. Understanding the purchasing behavior of different customer groups can help businesses improve loyalty programs and customer retention strategies.

5. Shipping Preferences

The analysis of Standard and Express Shipping methods revealed differences in customer purchasing behavior. These findings can support better logistics planning and improve the overall customer shopping experience.

6. Discount Effectiveness

The study of discount usage across different products highlighted how promotional offers influence customer purchasing decisions. This information can help businesses optimize future discount campaigns.

7. Customer Segmentation

Customers were categorized into New, Returning, and Loyal segments based on their previous purchase history. This segmentation enables businesses to design personalized marketing campaigns and strengthen customer relationships.

8. Product Demand Analysis

The analysis identified the most frequently purchased products within each category. Understanding product demand helps businesses optimize inventory management, reduce stock shortages, and improve sales performance.

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12. Business Recommendations

Based on the findings, the following recommendations are proposed:

- Strengthen customer loyalty programs to improve customer retention.
- Focus marketing efforts on high-value customer segments.
- Optimize inventory management for top-performing product categories.
- Develop personalized promotional campaigns based on customer demographics.
- Use interactive dashboards to monitor business performance regularly.

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13. Conclusion

This project successfully demonstrated the complete data analytics lifecycle using Python, PostgreSQL, and Power BI. Beginning with raw customer shopping data, the project involved data cleaning, exploratory analysis, SQL-based business analysis, and dashboard development.

The project highlights how data analytics can transform raw business data into meaningful insights that support strategic decision-making. It also provided practical experience with industry-standard analytics tools and strengthened my understanding of data cleaning, SQL, visualization, and business intelligence.

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14. Future Scope

The project can be extended further by:

- Developing customer segmentation using machine learning algorithms.
- Building predictive models for customer purchase behavior.
- Creating sales forecasting models.
- Publishing dashboards using Power BI Service.
- Integrating real-time retail data for continuous analysis.

15. References

1. Retail Customer Shopping Dataset (GitHub Repository)
2. Python Documentation
3. Pandas Documentation
4. PostgreSQL Documentation
5. Microsoft Power BI Documentation